

MACRAScript: A LANGUAGE FOR MACRAMÉ PATTERN

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Introduction

Macramé is a textile art form based on knotting, but its design process has traditionally lacked specialized digital tools, relying on manual, error-prone methods. Existing graphic design software is not optimized for the constraints of knot-based textiles. MacraScript is a Domain-Specific Language (DSL) created to bridge this gap. It provides a simple, declarative syntax for artisans, designers, and hobbyists to define macramé patterns programmatically, verify their structure, and generate clear visual guides, thus modernizing a traditional craft.

Goal

The main goal of this work is to design and implement a DSL, MacraScript, capable of simplifying the creation of complex macramé patterns. The research question is: *How can a DSL abstract the complexities of macramé to provide a user-friendly tool for pattern design and visualization?* The expected final product is a compiler that translates MacraScript code into dual graphical outputs: a final pattern preview and a step-by-step knotting diagram.

Proposed Solution: MacraScript Language and Compiler

We developed an external, declarative DSL with a simple grammar. The language is built around a **PATTERN** block, which can be **ALPHA** (for grid-based images) or **NORMAL** (for geometric knot patterns). The key innovation is the abstraction of knot-pairing logic in Normal patterns; the user only specifies the knot direction (**LEFT/RIGHT**), and the compiler handles the complex staggered thread pairing.

The compiler, written in Python, follows a traditional pipeline:

- **Lexical Analysis:** Converts code into tokens using regular expressions.
- **Syntactic Analysis:** A recursive descent parser validates the token stream against the grammar.
- **Semantic Analysis:** Validates logical rules and translates the abstract pattern definition into concrete knot instructions.

Conclusions

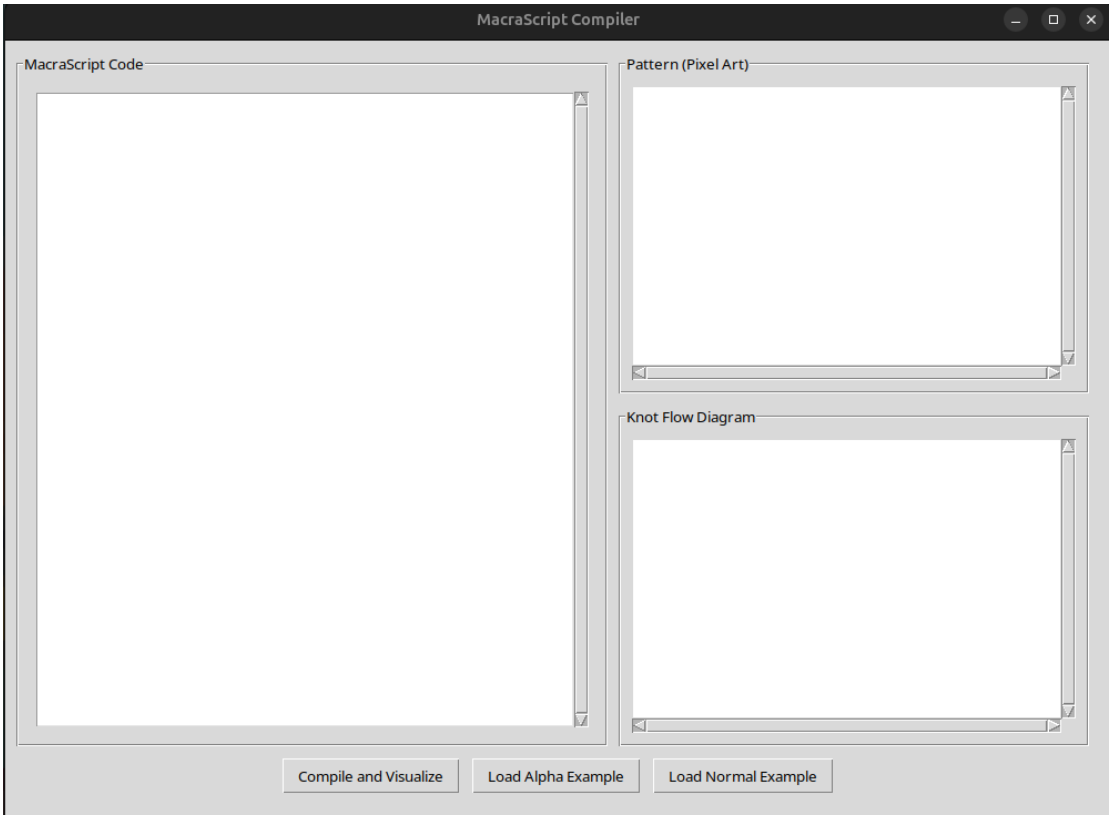
MacraScript successfully demonstrates that a DSL can effectively abstract the complexities of a traditional craft. By providing a simple declarative language, a robust compiler, and dual visualizations, we have created a powerful tool that lowers the barrier to entry for creating complex designs and reduces manual error. The project successfully answers the research question by providing a functional bridge between computational thinking and macramé art.

Bibliography

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3. Karner, C. (2005). *Macramé: The craft of knotting*. Schiffer Publishing.

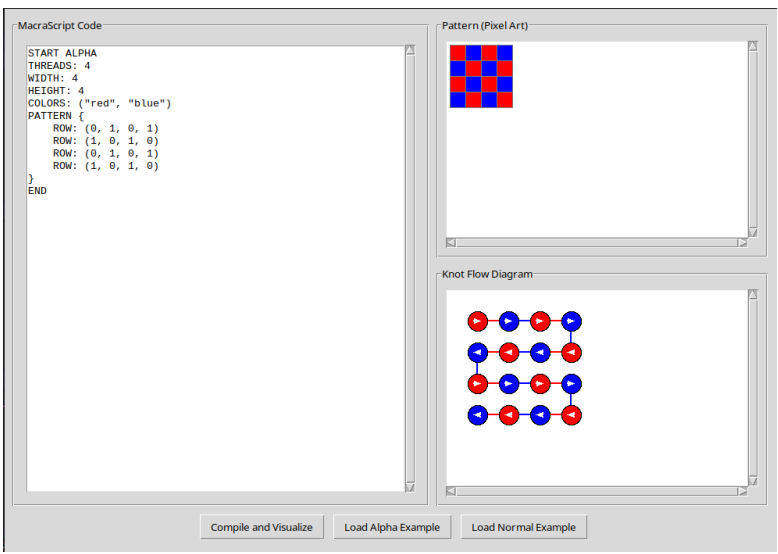
Visual Outputs

The compiler generates two distinct visualizations in a Tkinter-based GUI, serving both the design and execution phases of a project.

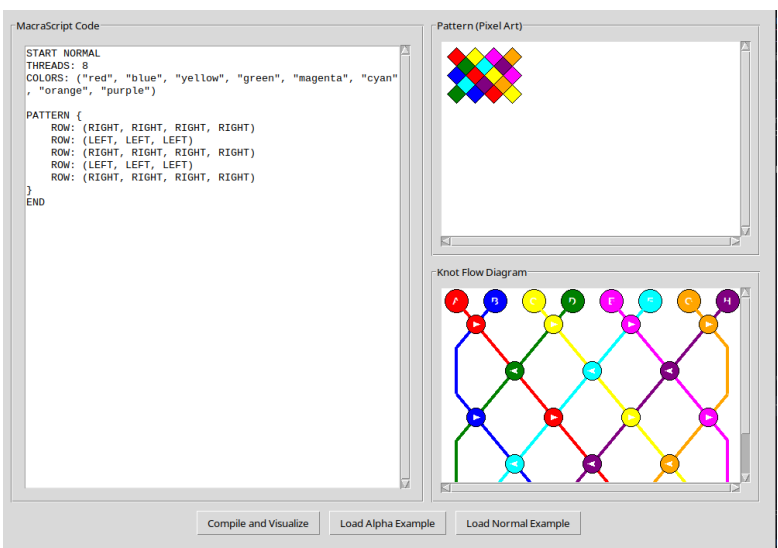


Experiments and Results

The language was tested with two main types of patterns to validate the compiler and visualization engine. The results confirm that the compiler correctly generates both a final preview and a knot-flow diagram for each pattern type.



Alpha Pattern Output



Normal Pattern Output